

# **Réalité virtuelle, environnements augmentés et objets connectés**

Le monde réel devient plus connecté que jamais

**Une présentation de  
Frédéric Guarino et Jonathan Bélisle**



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[iotheatre.co](http://iotheatre.co)

## Jonathan Bélisle

Entrepreneur web depuis plus de 20 ans, poète-réalisateur interactif, scénariste UX passionné, technologue créatif, professeur inspiré, Jonathan a étudié la narratologie au cinéma et a maîtrisé la pensée visuelle afin de les fusionner avec des méthodes d'ethnographie et de Design de services. Il approche de manière ludique les problèmes complexes reliés aux expériences connectées impliquant autant les univers narratifs, la domotique, les installations immersives, l'internet des objets, les médias en réseaux et les applications mobiles.

En tant que poète organisationnel et architecte d'entreprise, il a notamment supporté la transformation digitale de plusieurs compagnies. Fondateur de Hello, Architekt ! un studio de consultation en créativité organisationnelle et stratégies d'innovation, créateur de l'univers de Wuxia le renard, chercheur affilié au Annenberg Innovation Lab du Southern California University, professeur d'architecture d'expériences à l'INIS et Chef du Design chez SAGA où il a créé le premier système d'exploitation pour le monde physique.

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Jonathan Bélisle has been an Interactions Poet & Web entrepreneur for 20 years. As a polymath, he blends together his skills as an interactive designer, User Research Director, creative technologist and an inspired teacher.

He studied film narratology and mastered visual thinking before combining them with ethnographic methods and service design thinking. His diverse background enables him to approach complex problems in a playful manner when adding Human Touch to mobile devices and interactive storytelling to connected objects and media.

He is the founder of Hello, Architekt ! a Sensemaking & Changemaking Consultancy that helped the digital transformation of many enterprises, creator of Wuxia the Fox, a Research Affiliate at the Annenberg Innovation Lab of Southern California University, a professor of experience architecture at INIS Media School and Chief Design Officer at SAGA where he created the first OS for the Physical World.



## **Frédéric Guarino**

Spécialités: stratégie, développement des affaires

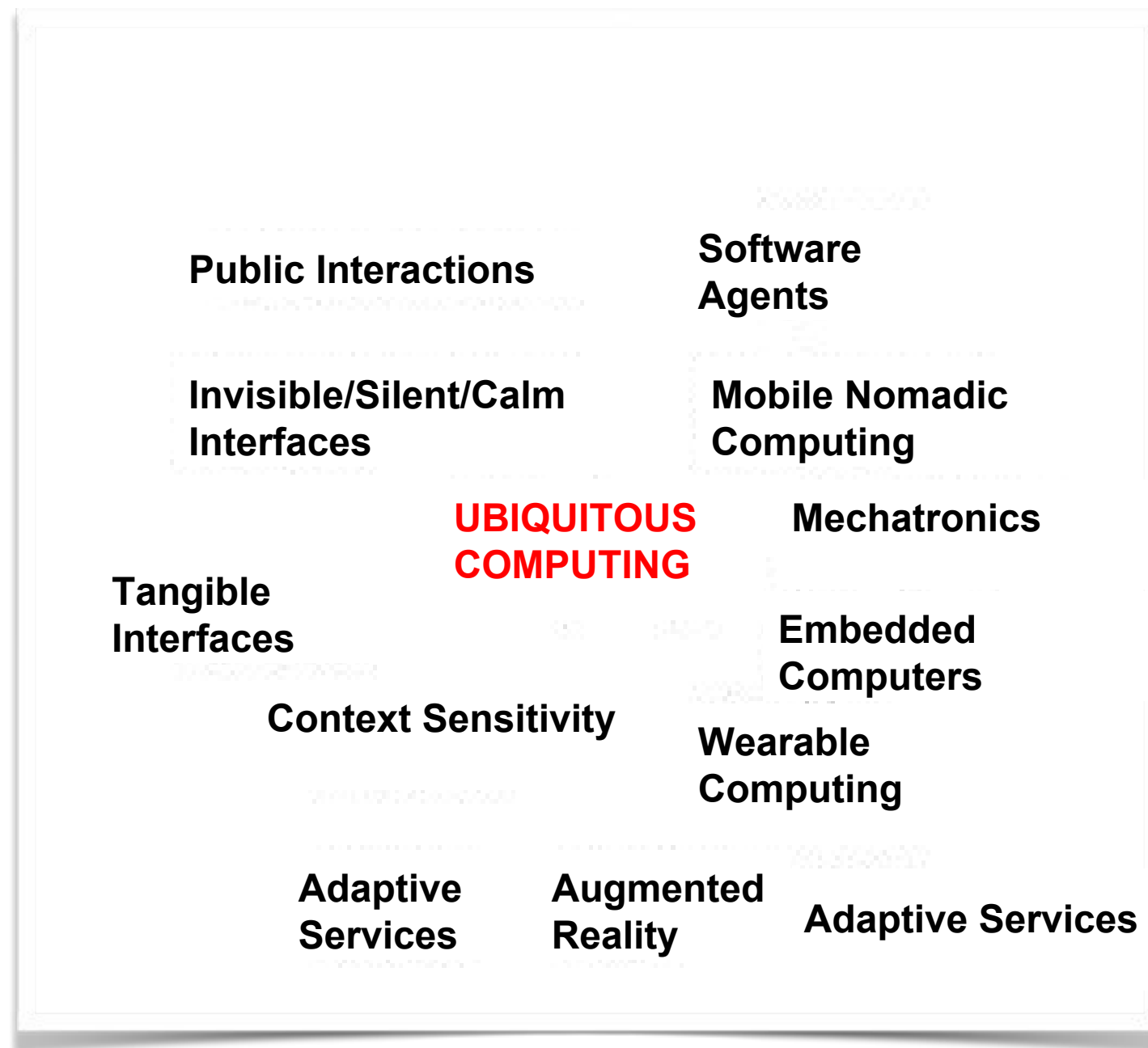
+18 ans à l'intersection des media et de la technologie, en développement des affaires, gestion de produit et stratégie pour des grands groupes (Lagardère New York), agence publicité/marketing (Hémisphère Group - Paris&New York), startups (Isentium - Montréal, Rixxo - Zurich/Londres/Paris)

co-organisateur du VR Salon de Mutek (prochaine édition 31 mai 2016)

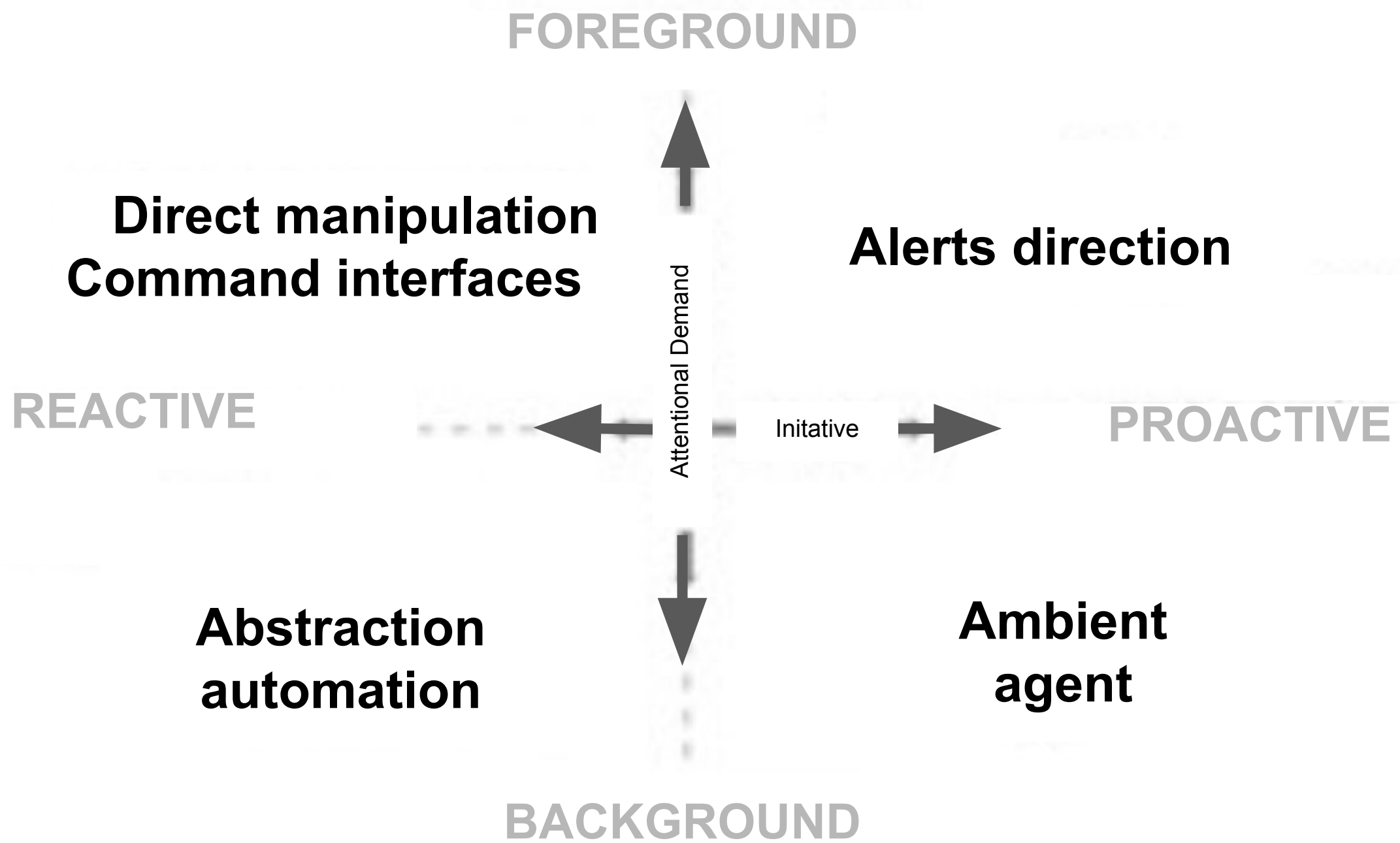
animateur de conférences personnalisées pour hauts dirigeants sur l'impact du numérique

# La prophétie de l'informatique calme

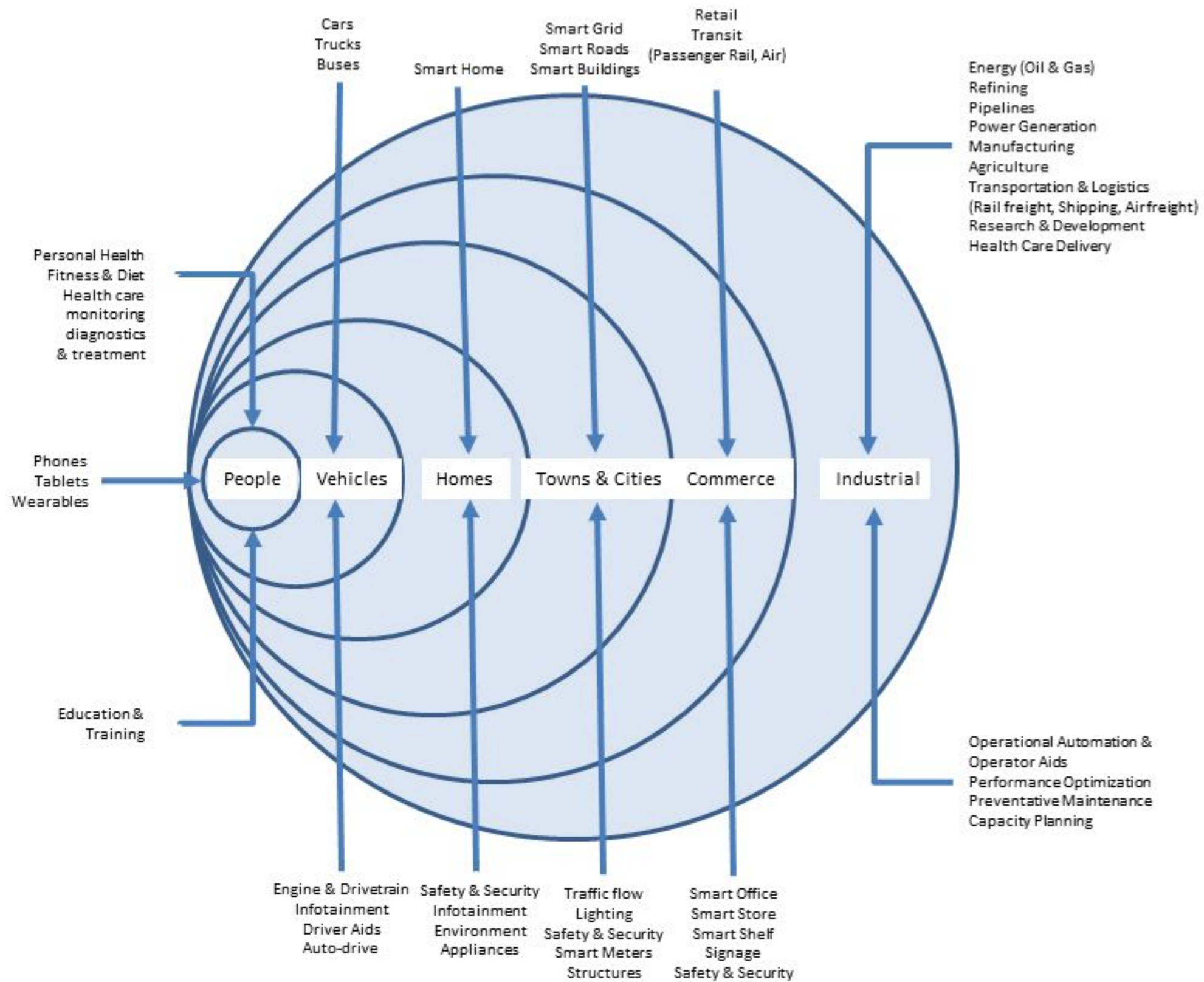
La rencontre de l'internet industriel et des réalités mixtes



# L'informatique calme - 1995



# L'internet des objets - 2016

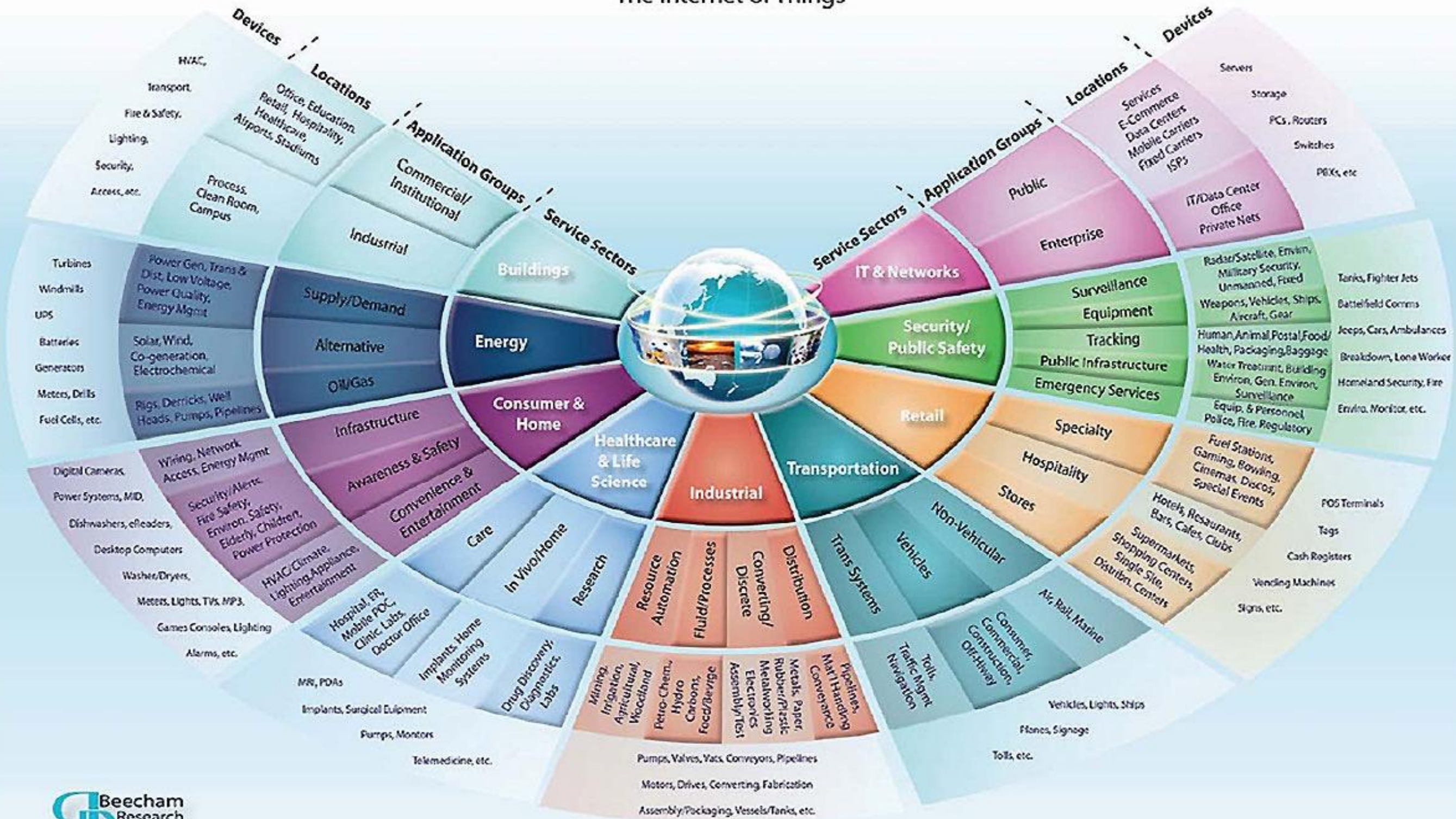


Source: Based on Goldman Sachs Global Investment Research. Additional analysis by WMG

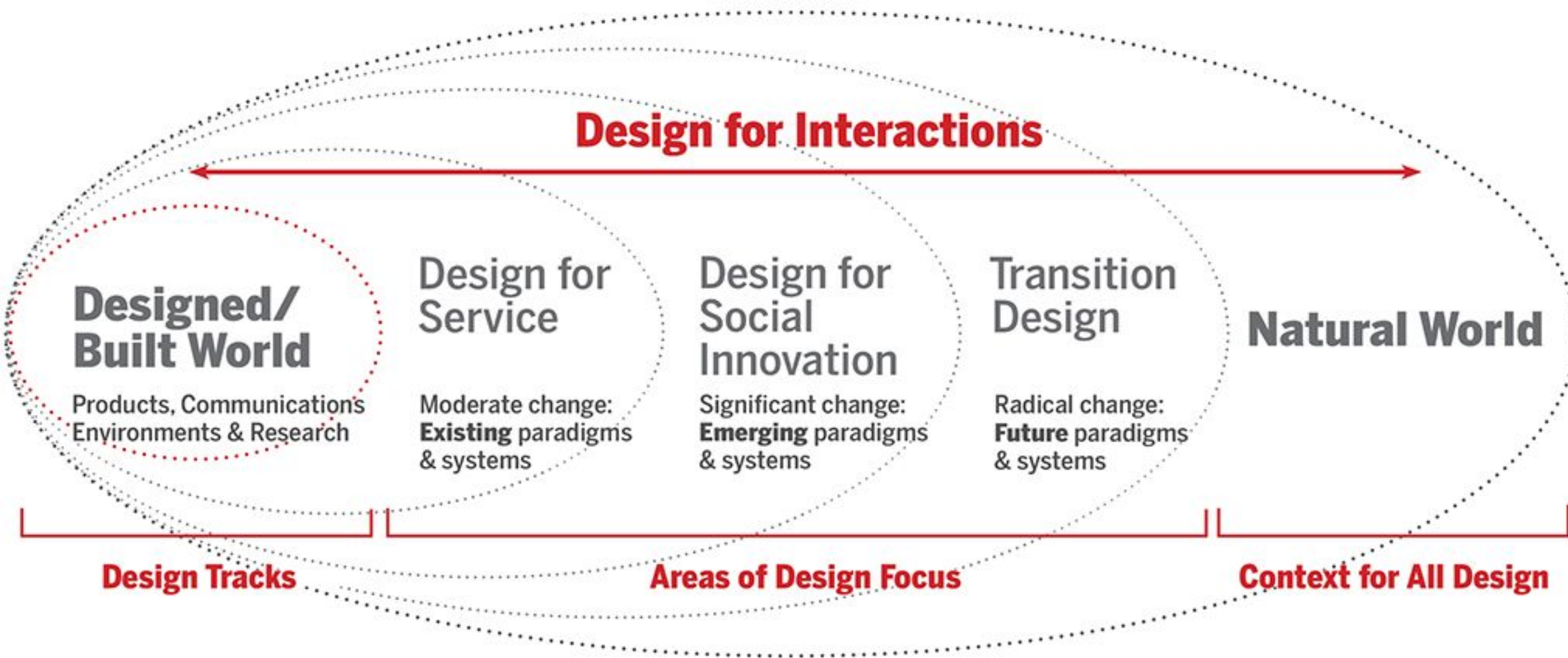
# IIoT et IoT - 2016



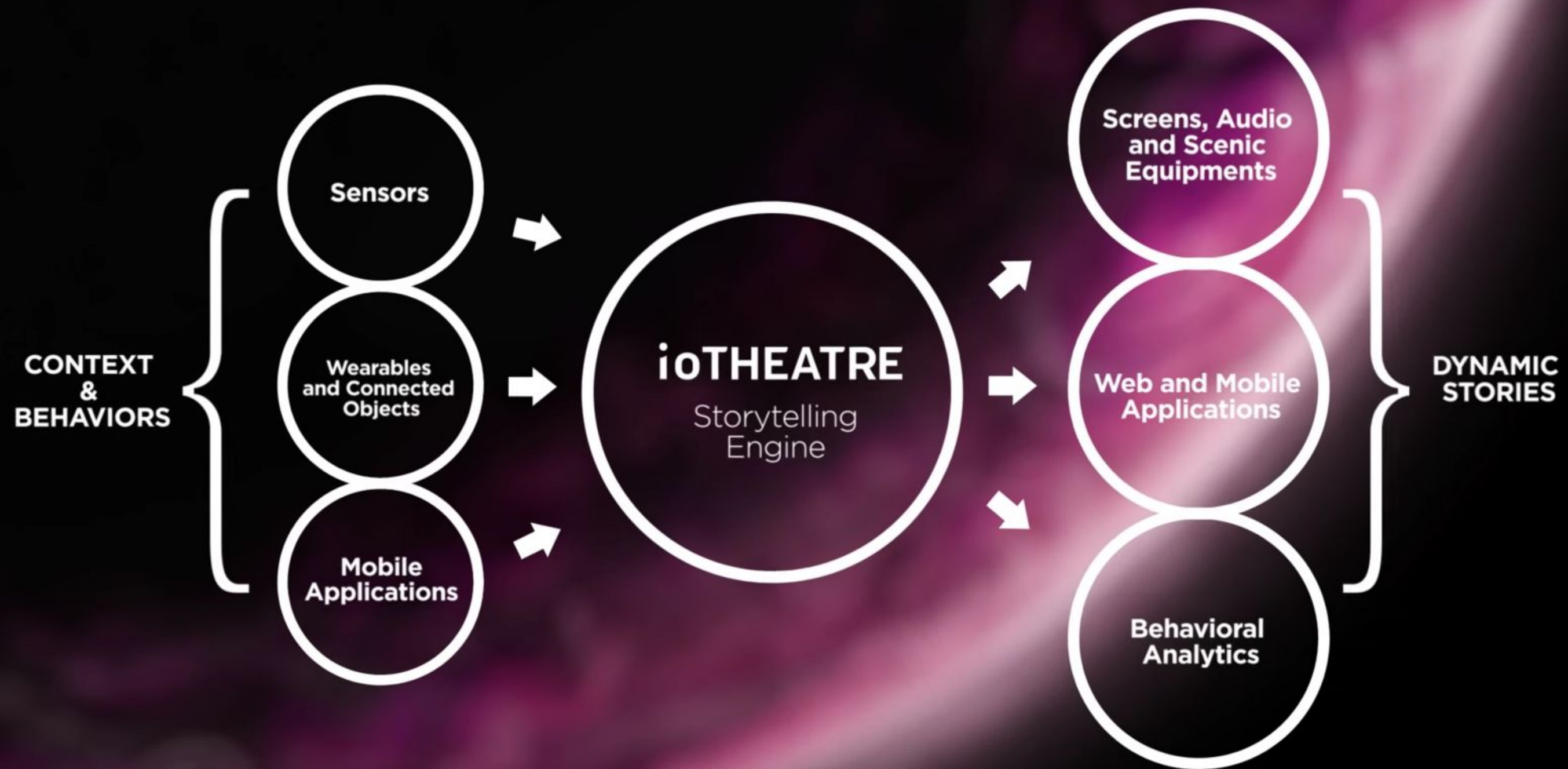
## The Internet of Things



# Design d'interactions - 2016



# Plateforme de Design d'expériences IoT - 2016



# Plateforme de Design d'expériences IoT - 2016

RITUELS

Interactions MapTimelineAssets Library

▼ Maps

● Geo Map

▼ Scenes

● ePaperDisplay

● Projection murale

▼ Inputs

● Casting01

● PeopleCounter

● ePaperDisplayRFID

● Input-test

▼ Outputs

● philtest

● ePaperDisplay

MapSatelliteGoogle

Map data ©2016 GoogleTerms of UseReport a map error

Scene

ePaperDisplay

Map: Geo Map

EditDeleteAdd next

Scene events

Add a new event

Kamisoris  
2 actions, 1 rules

LameDroite  
2 actions, 1 rules

Mach3  
2 actions, 1 rules

Surete  
2 actions, 1 rules

I/O DevicesAnalyticsjonathan@sagaworld.caLogout

Kamisoris

Trigger actions

IfThenSummary

Edit actions timing in Timeline mode

1520Raw data

KeyValue

itemA3.BMP

☐ Play after the previous asset has finished

☐ If another asset is playing, ignore this one

☐ Play while nothing is being played on the device

Output device: ePaperDisplay

1522Raw data

KeyValue

itemB4.BMP

☐ Play after the previous asset has finished

☐ If another asset is playing, ignore this one

☐ Play while nothing is being played on the device

Output device: ePaperDisplay

# Plateforme d'analyse spatiale - 2016



# L'analyse spatiale et le Sensor Fusion pour orchestrer les expériences AR/VR

Des marqueurs physiques aux senseurs en réseau



# VR DE MASSE MAINTENANT = CARDBOARDS GRIFFES – CAMPAGNE PRINT 2.0

En utilisant les téléphones intelligents, griffer les Cardboards aux couleurs des marques est un moyen excellent de massifier une campagne VR

- Facile à distribuer
- Coût relativement faible
- Bonne qualité
- Un “branded goodie”
- Comme une campagne imprimée

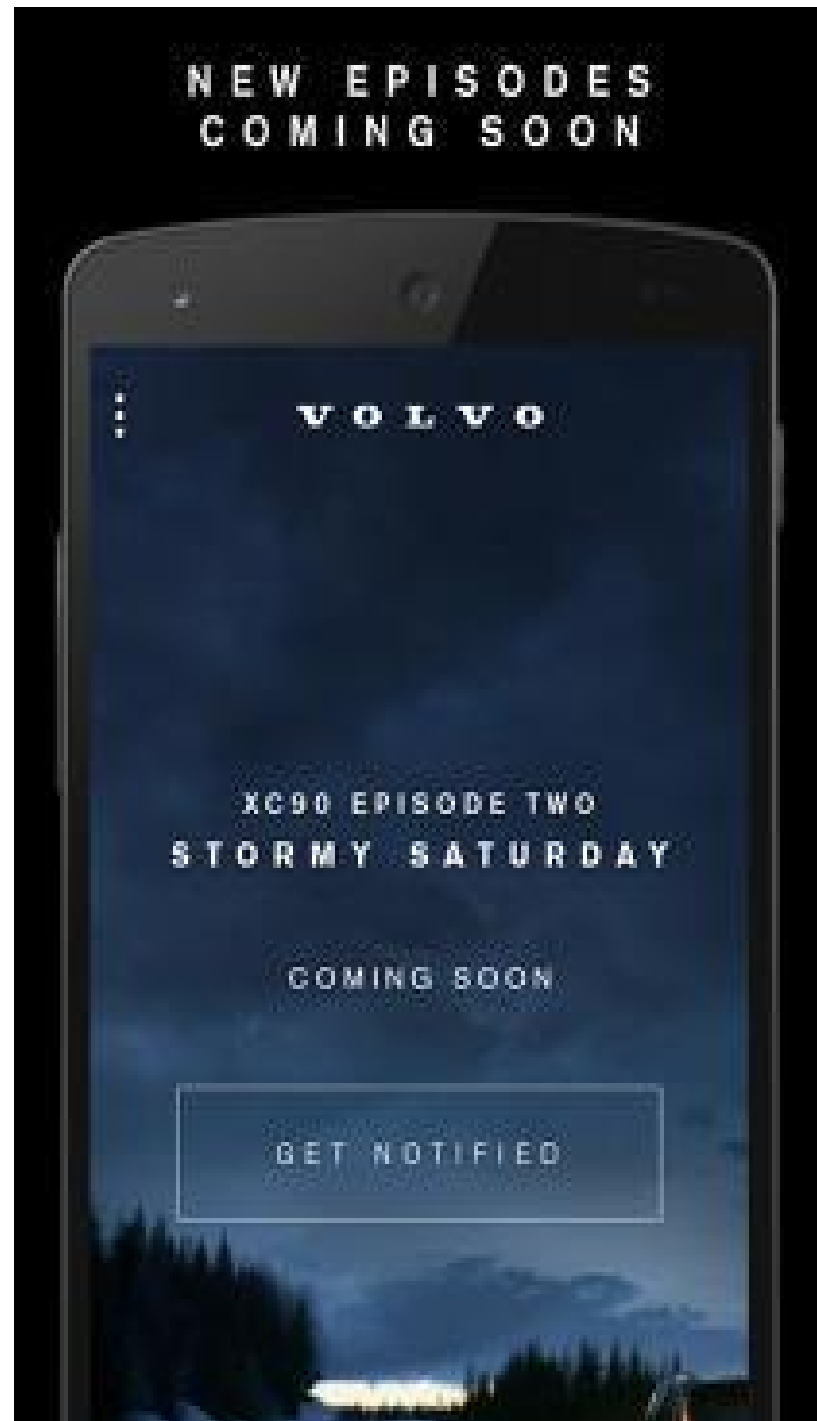


## LOISIRS, TOURISME & VOYAGE: MARRIOTT



Agence: Relevent NYC

# AUTOMOBILE: VOLVO



Agence: R/GA

# LE NY TIMES A DISTRIBUE 1M DE GOOGLE CARDBOARDS

Ben Landis follows

**Aaron Schmidt** @byaaron Schmidt · Nov 8

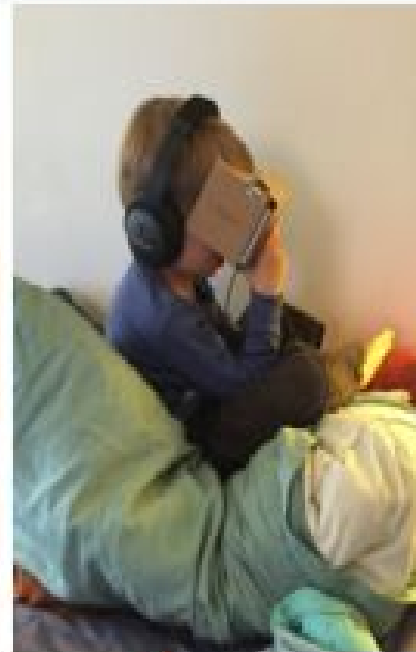
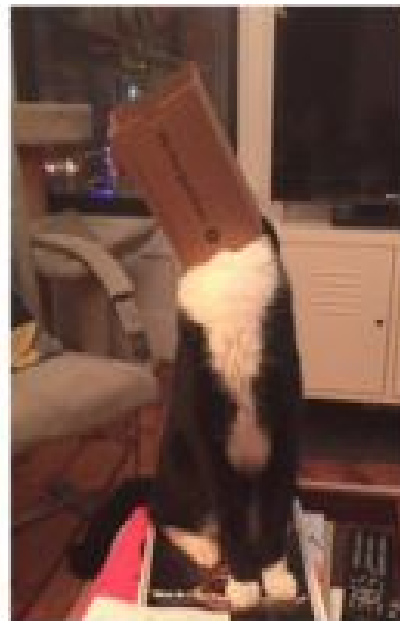
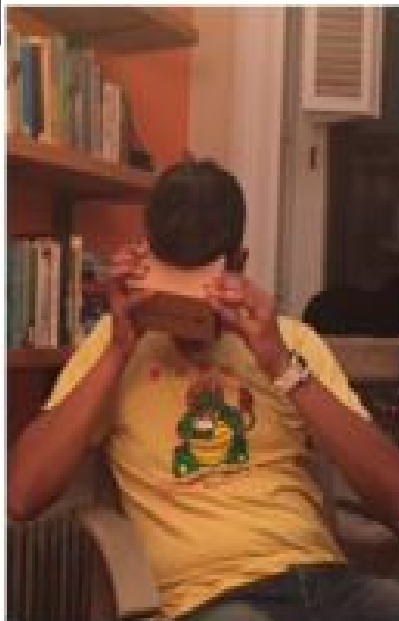
@nytimes @nytmag My five-year-old is reading his first New York Times.  
#NYTVR



Ad Age | daily

## How Mini, GE, Google and The NY Times Created a Watershed Moment for Virtual Reality

The Times' VR app was downloaded more times in its first four days than any Times app before it.



# LA VR MOBILE VA DOMINER EN 2016-2017

La VR mobile avec Samsung Gear VR, Google Cardboard mais aussi le Zeiss VR One permettent à tous les utilisateurs de téléphones intelligents de plonger dans la réalité virtuelle.

Les casques Oculus, HTC Vive et Playstation VR vont plaire aux “gamers” mais resteront coûteux.

## Samsung Gear VR sells out on Amazon and BestBuy.com

Christina Medici Scolaro

1 Hour Ago



# LA VR VA REINVENTER LE DIVERTISSEMENT

Le parc The Void en Utah montre la voie quant aux possibilités offertes par la VR pour réinventer le divertissement.



## 'These Guys Created a Real-life Holodeck ... Here's What it's Like to Use'

Time Magazine took a look at the Void and their article today shows they liked what they saw:

The future of virtual reality is looking more promising than ever. Now a Utah-based startup is hoping a project dubbed The Void will bring the visions of Star Trek, The Matrix, and William Gibson's Neuromancer one step closer to reality.

The equipment and technology required to make this dream a reality is designed to say out of the way of the experience.

"The transition from the real world to The Void is natural because the equipment is your size and comfortable and once you done your rig, the VR room is identical to the real-world room you've geared up in," Bretschneider says. "We want customers to forget about the equipment and enjoy the experience. It's about coming back again and again just like you enjoy new movies at the IMAX."



# VR landscape

## “Fixed” VR



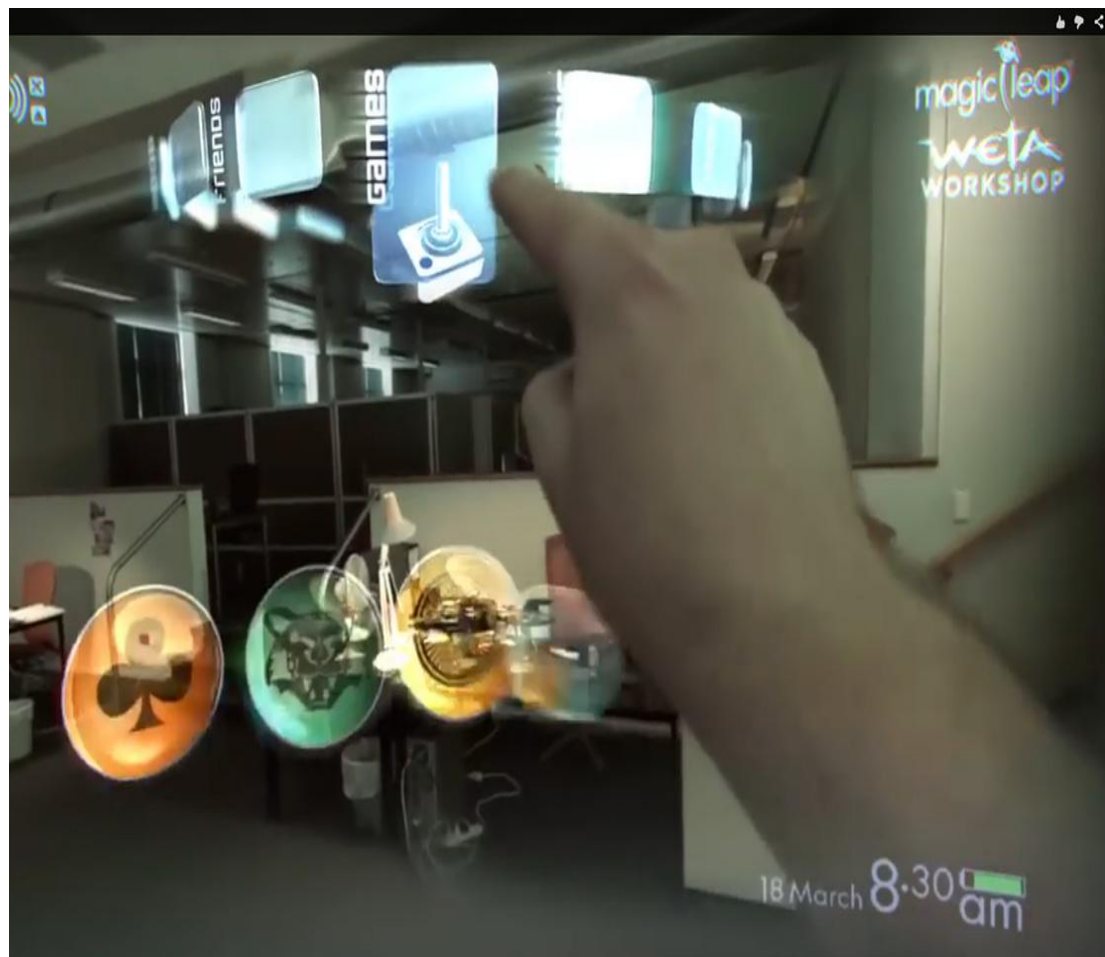
## Mobile VR



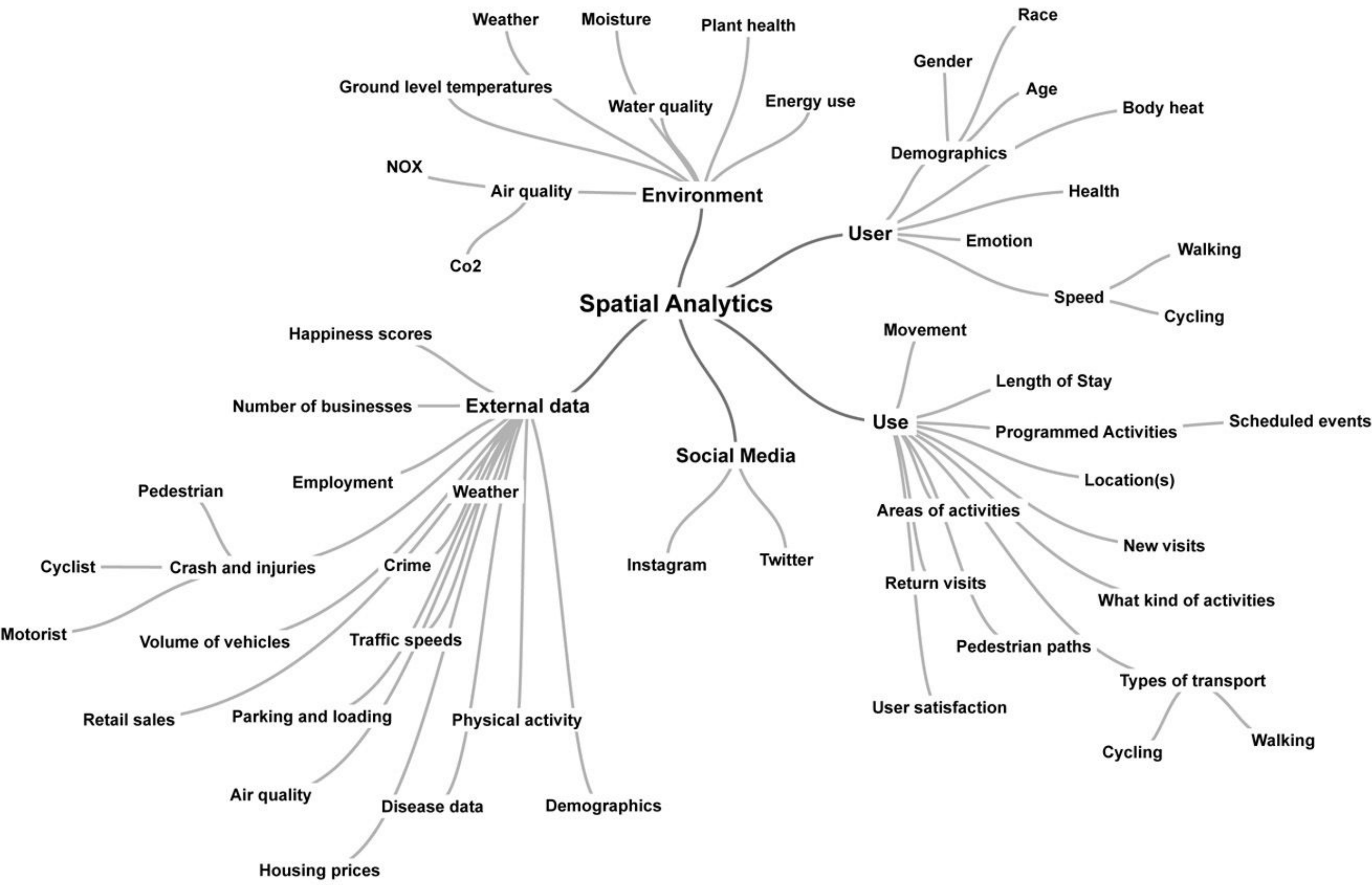
## AR landscape

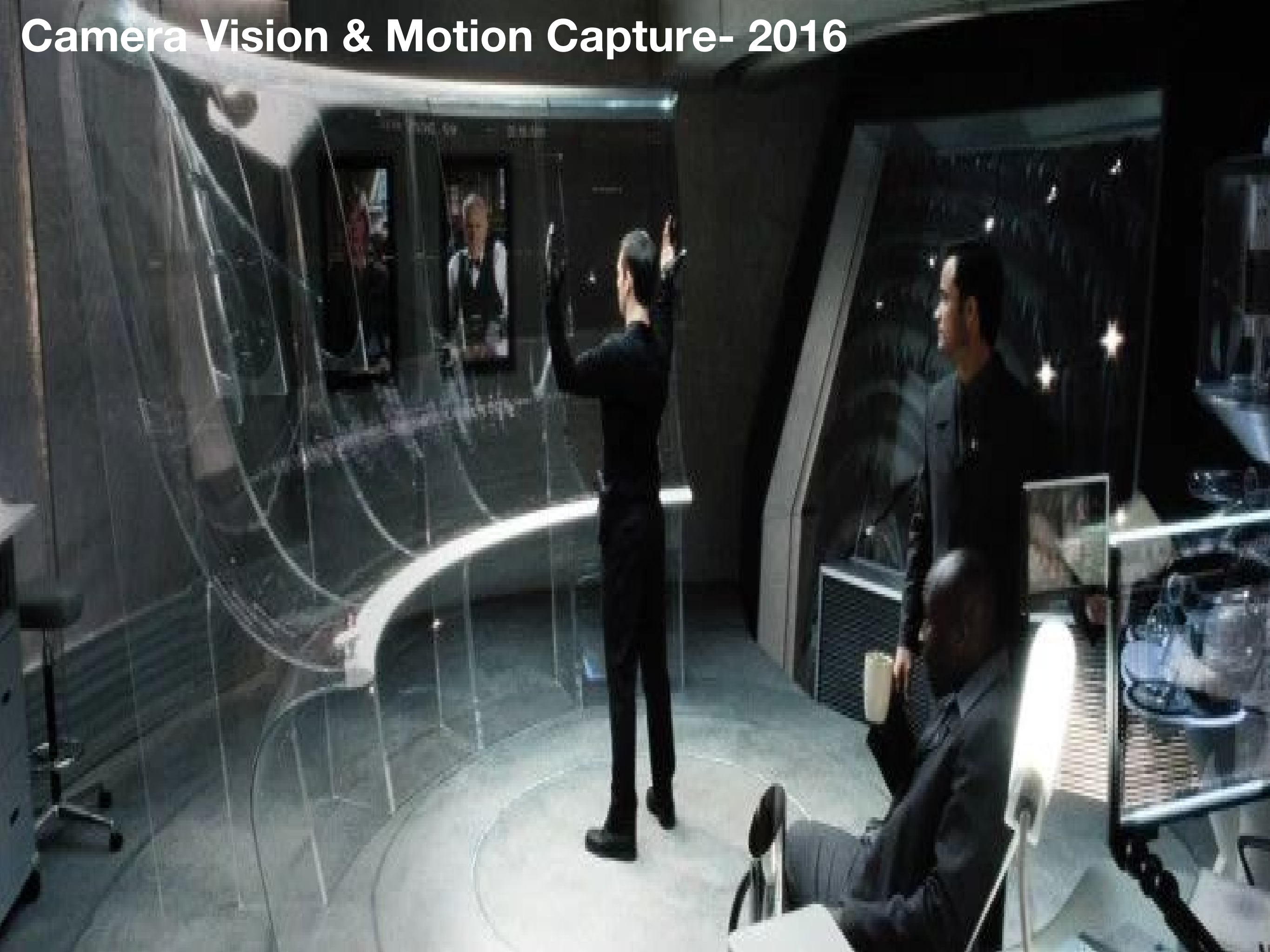


## AR landscape



# Analyse spatiale - 2016

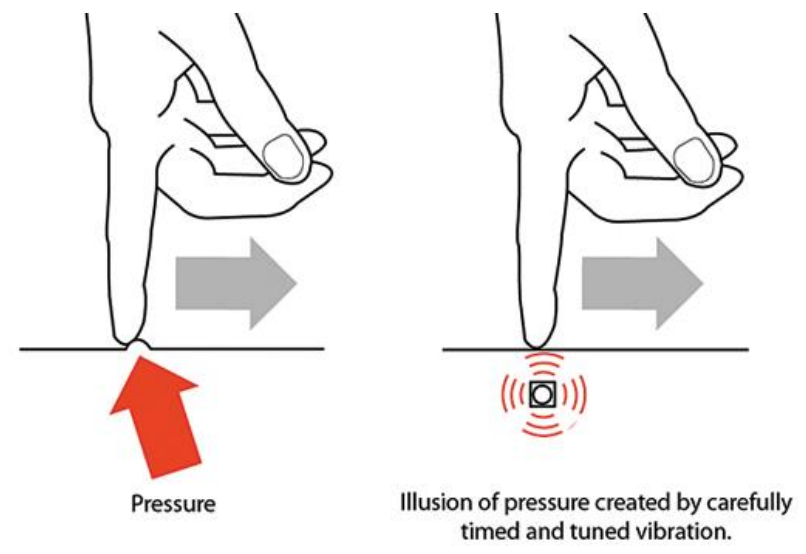
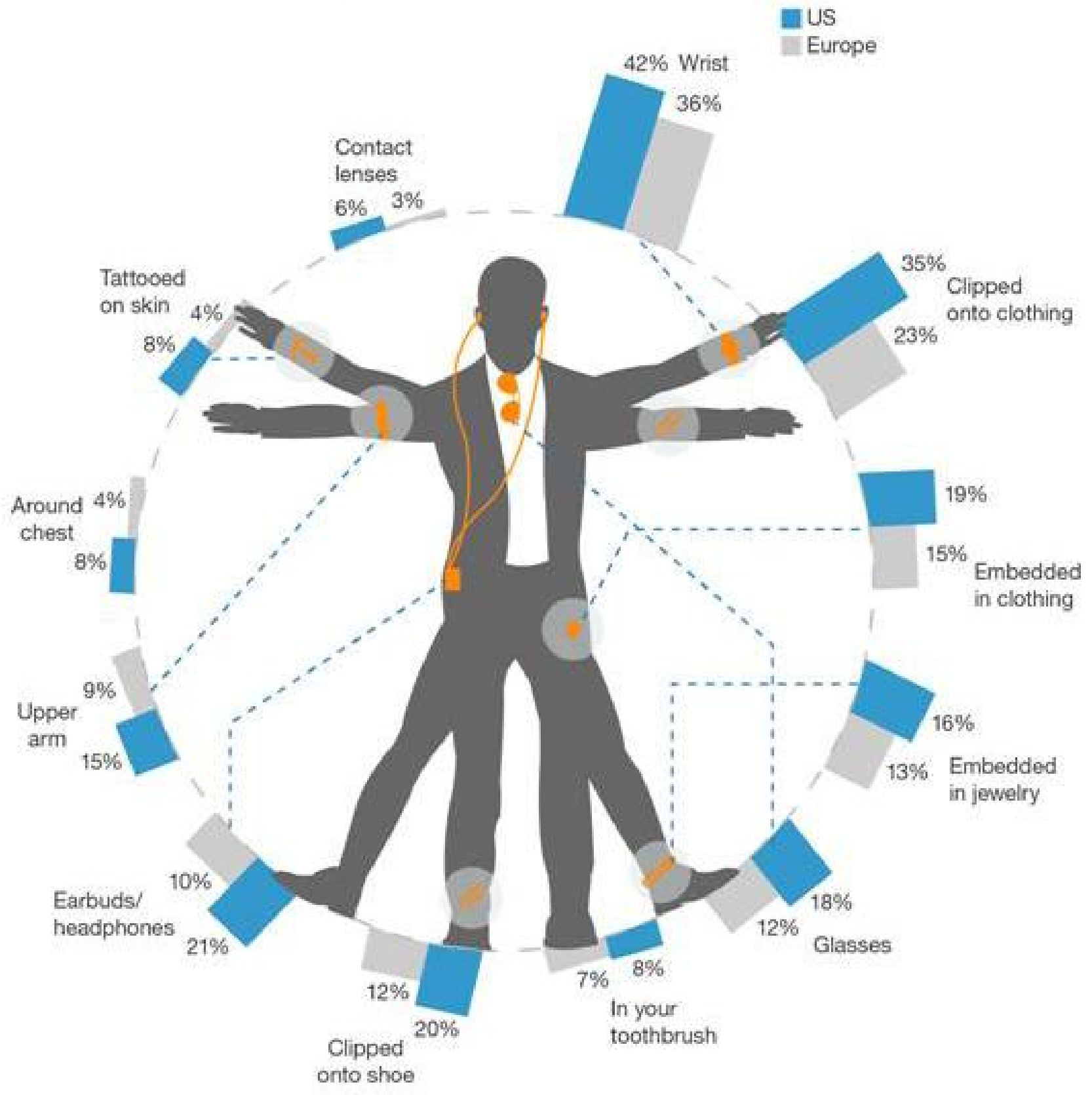


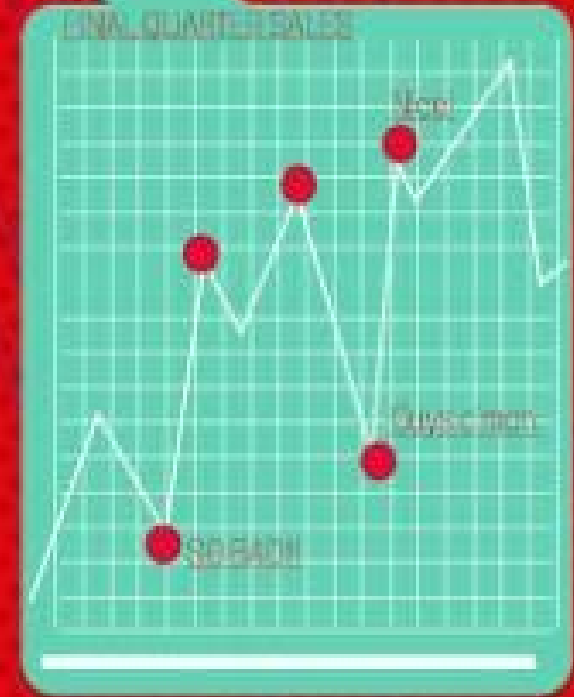


# Camera Vision & Motion Capture- 2016

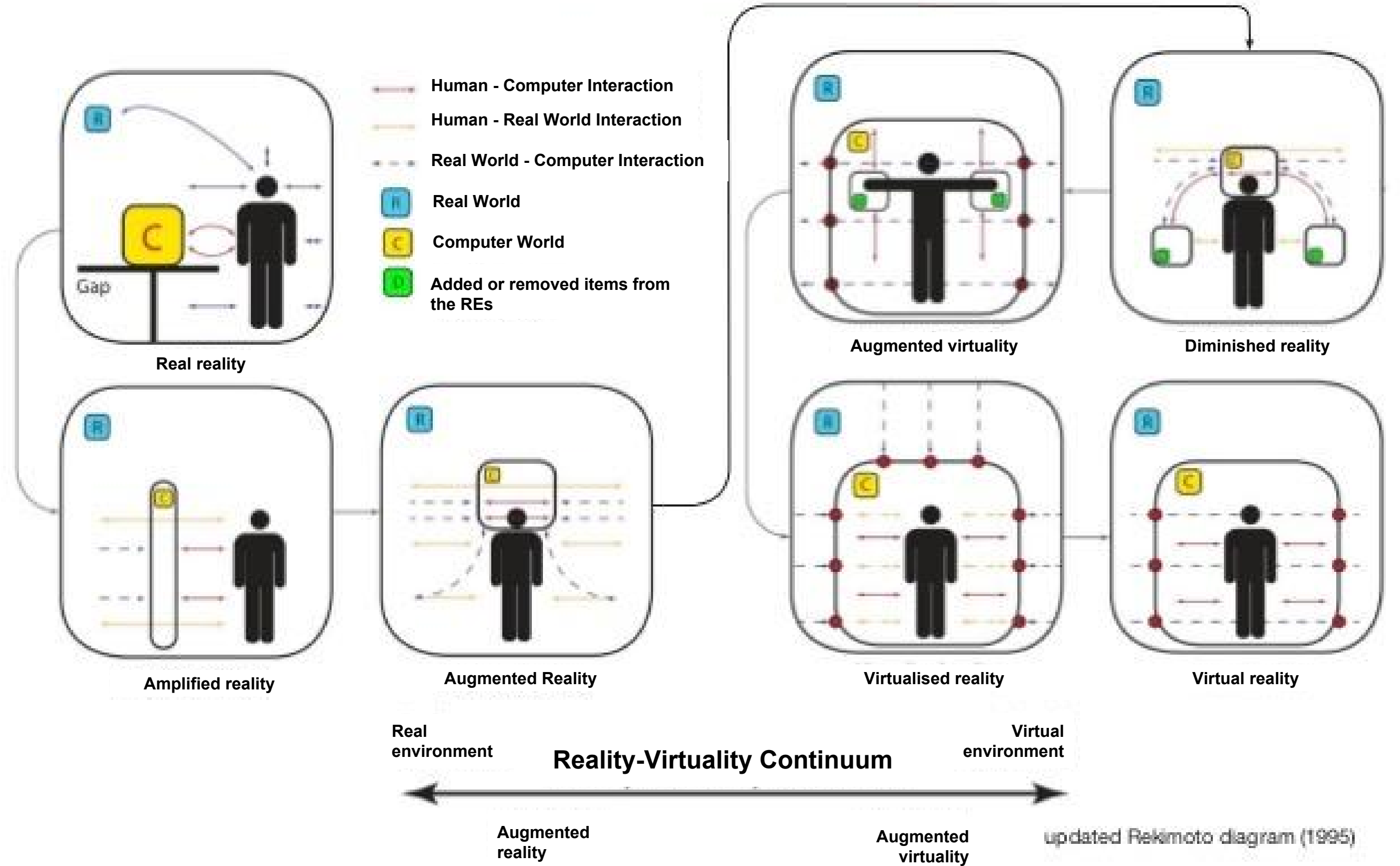
# Interfaces Haptiques et autres senseurs portables - 2016

"How interested would you be in wearing/using a sensor device, assuming it was from a brand you trust or offered a service that interests you?"





# Réalités mixtes - 2016



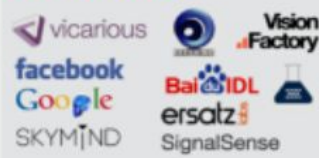
# Intelligence machine - 2016

## CORE TECHNOLOGIES

### ARTIFICIAL INTELLIGENCE



### DEEP LEARNING



### MACHINE LEARNING



### NLP PLATFORMS



### PREDICTIVE APIS



### IMAGE RECOGNITION



### SPEECH RECOGNITION



## RETHINKING ENTERPRISE

### SALES



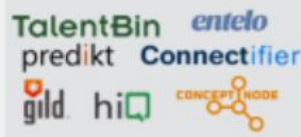
### SECURITY / AUTHENTICATION



### FRAUD DETECTION



### HR / RECRUITING



### MARKETING



### PERSONAL ASSISTANT



### INTELLIGENCE TOOLS



## RETHINKING INDUSTRIES

### ADTECH



### AGRICULTURE



### EDUCATION



### FINANCE



### LEGAL



### MANUFACTURING



### MEDICAL



### OIL AND GAS



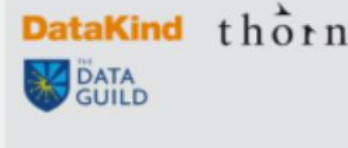
### MEDIA / CONTENT



### CONSUMER FINANCE



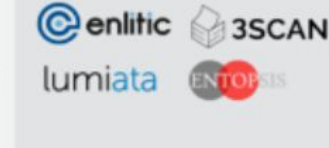
### PHILANTHROPIES



### AUTOMOTIVE



### DIAGNOSTICS



### RETAIL



## RETHINKING HUMANS / HCI

### AUGMENTED REALITY



### GESTURAL COMPUTING



### ROBOTICS



### EMOTIONAL RECOGNITION



## SUPPORTING TECHNOLOGIES

### HARDWARE



### DATA PREP

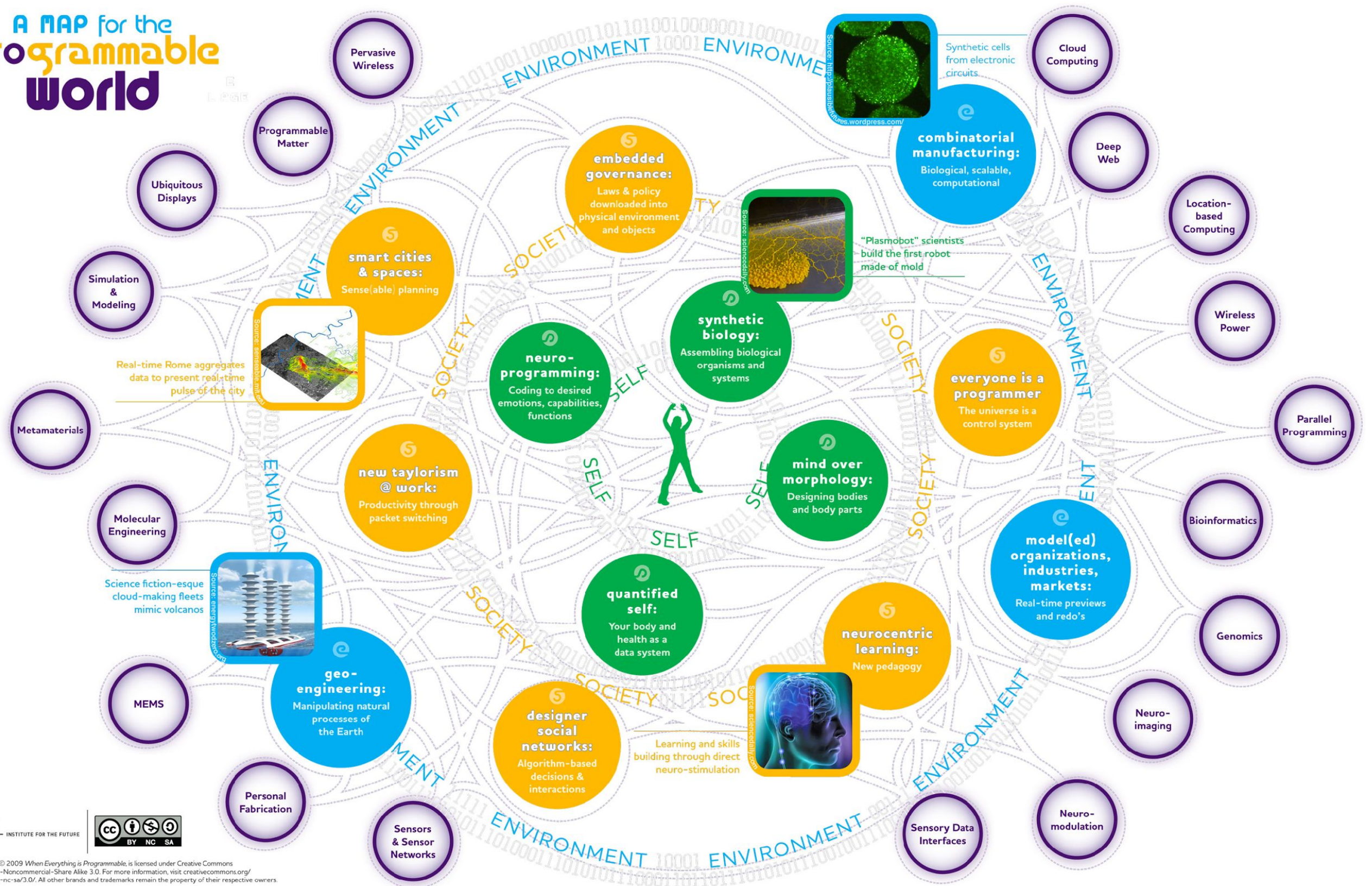


### DATA COLLECTION



# Vers un monde programmable

## A MAP for the programmable world



# ROBOT RENAISSANCE

## THE RIGHT TOOL FOR THE JOB

From a global division of labor to a robotic workforce

White collar robots

Rise of the robot scientists

Armies become autonomous

"Klaatu barada nikto!"  
- The Day The Earth Stood Still (1951)

## PHYSICALIZING THE DIGITAL

From virtual to blended reality

Interfaces become tangible

Every human-machine interaction is a data point

Robots are media

"Thank the maker!"  
- C-3PO, Star Wars Episode IV: A New Hope (1977)

## ROBOTS IN THE WILD

From rigid environments to dynamic infrastructures

Networks are interoperable

Systems self manage

The world is tagged

## THE FUTURE OF HUMAN-MACHINE INTERACTION

## NEITHER ARTIFACT NOR LIVING BEING

From form factors to phenotypes

AI is rethought, again

The uncanny valley is elevated

A new category of being emerges

## OUR ROBOTS, OURSELVES

From mimicking humans to decoding our behaviors

Robot fetishism goes mainstream

Machines write a how-to guide for humans

## HUMAN-MACHINE MUTUAL AID SOCIETY

From executing programs to mutual self-improvement

Symbiosis guides evolution

Robots teach humans new tricks

Hideki Kozima's Keepon robot is designed to investigate the role of movement in communication and is being used in autism research.

"A robot may not injure a human being, or, through inaction, allow a human being to come to harm."

- Asimov's First Law of Robotics, *Runaround*, 1942

## AUTOMATE

## AUGMENT

## UNDERSTAND

## UNDERSTAND

## AUGMENT

## AUTOMATE



Systems like Contour Crafting, a construction technology developed at USC, promise to revolutionize production of physical buildings and infrastructure.



CONE is a hybrid teleoperated/autonomous robotic observatory for remote field biology, from UC Berkeley and Texas A&M University.



The introduction of robotic systems into combat is already leading some to speculate that these systems will catalyze a complete reevaluation of war.



Operating room robots like the da Vinci Surgical System provide human physicians with incredible levels of accuracy.

"I'm sorry I'm not real. If you let me, I'll be so real for you!"  
- David, android child, Steven Spielberg's A.I. (2001)



Osaka University's CB2 robot mimics behaviors of human infants.



The form factor and dexterity of NASA's Robonaut is intended to allow the system to work alongside human astronauts.



ATR and Systec Akazawa's Muu Socia 3.0 is a "communication support" for better physician/patient interaction.



Nexi is a dexterous robot capable of a great range of expressiveness designed to more richly communicate stories to human users.



Google's project to automate vehicles, based on technologies developed for DARPA Challenge races, will literally put robotics in the driver's seat.



The HiBot is a Tokyo-based system designed to inspect multiple powerlines at once.



Tokyo University of Science's Saya is a robot teacher that expresses six basic emotions.



In his book *Love And Sex With Robots*, David Levy explores our deepening emotional attachment to machines.